Project plan : Parametric garments

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1.Project : Parametric garments (code: sewing-t-shirt-1)

A lightweight web tool that lets users tweak simple garment parameters (sleeve length/opening, torso length, neckline, fit) and then produces a live 3D preview and downloadable cut-ready sewing patterns (SVG/PDF).

See a Blender add-on which allows converting 3D models to sewing patterns: https://thomaskole.nl/s2s/

Objectives:

- Build a compact web UI to control garment parameters and preview results in 3D.
- Implement a backend hook that runs Blender headless to apply parameters and export sewing patterns (This step can be replaced by another solution or in-house built sewing pattern generator).
- Deliver printable SVG/PDF patterns scaled to basic measurements and a sewn prototype demonstrating fit. Sewing machine available via ESIEE Clubs

1. Project context and goal

We are implementing the “Parametric garments (sewing-t-shirt-1)” project. Github link : Github

Goal: build a lightweight web tool that allows users to:

- Adjust simple garment parameters (torso length, sleeve length, sleeve opening, neckline, fit) this in case we do a T-shirt.
- See a real-time 3D preview of the garment.
- Generate and export cut-ready sewing patterns in SVG and PDF, at true scale.

We chose to implement an **in-house sewing pattern generator in JavaScript** instead of a Blender headless backend.

2. Functional scope (what the app does)

1. Web UI
2. Real-time 3D preview
    - 3D view of a stylized garment (torso + sleeves) in a canvas.
    - Orbit camera (rotate and zoom with mouse).
    - Mesh automatically recomputed whenever a parameter changes.
3. 2D pattern generation
    - Three pattern pieces:
        - Front body
        - Back body
        - Sleeve
        - …
4. SVG and PDF export at scale
    - SVG:
        - White background, main outline, dashed seam allowance outline.
    - PDF:
        - A4 landscape format.
        - Automatic tiling when a piece is larger than a single page, with overlap and registration marks.
        - Header text reminding to print at 100% (no scaling).
    - Files are ready to print on standard A4 paper.
5. Sewn prototype

- At the end of the project, we will print a generated pattern and sew a small prototype (simple t-shirt or smaller accessory) to demonstrate that the pattern is usable.

3. Tech stack and tools

- Languages:
  - HTML5
  - CSS3
  - JavaScript (ES6 modules)
- Libraries:
  - Three.js
    - 3D scene, camera, lights, garment mesh (torso + sleeves).
    - OrbitControls for interactive camera navigation.
  - SVG.js
    - Drawing sewing patterns as SVG from centimetre coordinates.
  - jsPDF
    - Generating A4 PDFs, tiling, registration marks, and scale bar.
- Important design choice:
  - We do not use Blender in the runtime pipeline.
  - Instead, we rely on a custom JavaScript pattern generator, which matches the “in-house built sewing pattern generator” option described in the project specification.

4. Work organization (task split and timeline)

- Week 1
  - Set up project structure and dark-theme UI.
  - Implement parameter store and controls.
  - First working version of the 3D preview.
- Week 2
  - Complete pattern generator.
  - Implement working SVG export.
  - Refine garment 3D shape to look closer to a t-shirt.
- Week 3
  - Implement PDF export with A4 tiling and registration marks (jsPDF).
  - Print tests to validate scale (10 cm bar).
  - Final documentation (README).
  - Make a small physical prototype from one generated pattern.

Task split:

- **Hocine**: Web UI, parameter controls, Real-time 3D preview
- **Mohamed**: 2D pattern generation, mesh geometry
- **Rayan**: SVG and PDF export at scale, Sewn prototype and documentation.